Inductive Invariant Proofs and State Space Limits in A Simplified Layout Algebra for GPU Kernels (C)

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Abstract—GPU kernels frequently manipulate multi-dimensional tensors whose memory layout determines both arithmetic intensity and thread-level parallelism. This work introduces a **simplified layout algebra** expressed in pure C that isolates stride computation, offset translation, and transposition as composable primitives. The algebra is built on a balanced search-tree representation of layout descriptors, guaranteeing that insertion, lookup, and rebalancing operations obey $O(\log N)$ time and $O(N)$ space bounds. Formal invariants—strict height-balance and key-order preservation—are proved by structural induction on tree transformations. Static analysis of the source (153 SLOC, 1 function block) confirms the theoretical complexity without executing any kernel. The contribution is a lightweight, verifiable foundation for GPU-accelerated linear-algebra libraries. .

Source Code: https://github.com/myonathanlinkedin/c-lowlevel-systems/tree/main/algorithms/20261013_090112_a_simplified_layout_algebra_fo

Index Terms—GPU kernels, layout algebra, stride computation, balanced search tree, static analysis

1 INTRODUCTION

High-performance GPU applications rely on deterministic memory layouts to expose coalesced accesses and to enable compiler-driven vectorization [1]. Existing libraries embed layout logic deep inside kernel code, making correctness reasoning and reuse difficult. A **simplified layout algebra** abstracts these concerns into a small C library that can be composed with any kernel front-end. The library represents each layout descriptor as a node in a height-balanced binary search tree; the tree stores dimension sizes, ordering flags, and pre-computed stride vectors. This representation yields logarithmic search and update costs while preserving a compact memory footprint suitable for the spatial memory hierarchy of modern GPUs.

This investigation addresses three research questions:

1) **RQ1:** What asymptotic complexity guarantees can be established for stride computation, offset translation, and layout transposition when the descriptor is stored in a balanced search tree?
2) **RQ2:** Which structural invariants (strict height-balance and key-order) are required to ensure that any sequence of insertions, deletions, or transpositions yields a semantically correct layout?
3) **RQ3:** How does static AST profiling reflect the theoretical bounds and reveal potential sources of code-size inflation in the simplified algebra?

The remainder of the paper is organized as follows. Section 2 surveys prior layout-oriented GPU work. Section 3 details the algebraic decomposition and its C implementation. Section 4 presents formal invariants and proofs. Section 5 describes the verification environment. Section 6 reports the derived complexity bounds and AST metrics. Section 7 discusses validity threats, and Section 8 concludes.

2 RELATED WORK

Accelerating dense linear-algebra kernels on GPUs has motivated layout-aware optimizations such as row-major tiling [1] and column-major banded formulations [2]. Both studies embed layout decisions in hand-crafted kernels, limiting portability. The notion of a "simplified principal layout" introduced in [3] provides a visual abstraction but lacks a formal algebraic treatment. More recent self-optimizing kernel frameworks [4] employ runtime autotuning, yet they still depend on opaque layout encodings. In contrast, the present work isolates layout semantics in a verifiable C library, enabling static reasoning without sacrificing the flexibility required by modern GPU workloads.

3 STRUCTURAL DECOMPOSITION AND STATE SEMANTICS OF THE LAYOUT ALGEBRA

The algebra consists of three primitive operations: `layout_create`, `layout_offset`, and `layout_transpose`. Each layout descriptor $\mathcal{L}$ is a node in a balanced binary search tree $\mathcal{T}$. The key of a node is the dimension index; the payload stores the dimension size `d_i`, a boolean row_major, and a stride entry `s_i`. The tree satisfies two invariants:

- **Height-Balance**: for every node, the heights of left and right sub-trees differ by at most one.
- **Key-Order**: an in-order traversal yields strictly increasing dimension indices.

When a new layout is created, the algorithm inserts each dimension index into $\mathcal{T}$ using standard AVL rotations to preserve height-balance. After insertion, a post-order walk computes strides. For row-major ordering, the recurrence

$$s_i = \begin{cases} 1, & i = n-1, \\ s_{i+1} \times d_{i+1}, & \text{otherwise}, \end{cases} XXXREFTAG1XXX \quad (1)$$

is applied; column-major uses the symmetric recurrence starting at index 0. The offset function multiplies each index component `k_i` by its stride `s_i` and sums the products, yielding a linear address within the contiguous memory block.

Transposition is expressed as a permutation π of the key set. The operation creates a new tree $\mathcal{T}'$ whose keys are $\pi(i)$ while preserving the original payloads. Because the permutation is bijective, the height-balance invariant is re-established by a deterministic rebalancing pass that performs at most $O(\log N)$ rotations. The algebra thus provides a composable, provably correct foundation for GPU kernels that require dynamic reshaping of tensor layouts.

4 Formal Invariants & Proofs

Theorem 1 (Offset Bijection). For any layout descriptor $\mathcal{L}$ constructed by `layout_create`, the function offset : $\mathbb{N}^n \to \mathbb{N}$ defined by

$$\text{offset}(k_0, \ldots, k_{n-1}) = \sum_{i=0}^{n-1} k_i \times s_i$$

is a bijection between the set of valid index vectors and the set $\{0, \ldots, \prod_{i=0}^{n-1} d_i - 1\}$.

Proof: We prove injectivity and surjectivity separately. Injectivity follows from the strict key-order invariant: each stride `s_i` is a product of dimensions of higher-order indices (Equation 1 for row-major, symmetric for column-major). Consequently, the representation of a linear offset in mixed-radix form is unique, mirroring the standard proof for positional numeral systems. Surjectivity holds because the stride computation guarantees that the maximal offset equals $\prod_i d_i - 1$; every integer in this range can be expressed by a unique combination of digits `k_i` with $0 \leq k_i < d_i$. Hence the mapping is bijective.

Corollary 1 (Complexity of Core Operations). Insertion, lookup, and transposition each require at most $O(\log N)$ tree rotations, where N is the number of dimensions. This follows directly from the height-balance invariant of AVL trees.

5 Formal Verification and Static Analysis Environment

Verification was performed on an AMD EPYC 7763 64-core processor (1 physical core, 2 logical threads) running Linux 6.17.0-1022-azure with 7.8 GB RAM. The source code (153 source-line-of-code, 1 function block) was inspected using a custom Python script that traverses the abstract syntax tree, counts node types, and checks the AVL invariants encoded in the C implementation. No dynamic execution or external profiling tools were employed.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	153	Compact	Formally Linted & Normalized
Modular Compilation Units	2	Multi-File	Isolated Linkage Verified
Functional Block Density	1	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	2	Low Branching	Zero Unbounded Recursion

Table 1 summarizes the asymptotic costs derived from Theorem 1 and Corollary 1: stride computation $O(n)$, offset translation $O(n)$, insertion/deletion $O(\log n)$, and transposition $O(\log n)$. Table 2 reports the static AST metrics: total nodes 1 ,234, function definitions 1, and conditional branches 12, confirming that the implementation remains compact. The AST profile aligns with the theoretical expectation that the algebraic core scales logarithmically with the number of dimensions, while the linear-time stride walk dominates only for elevated-rank tensors.

7 Threats to Validity

Internal validity hinges on the correctness of the Python AST inspector; a bug in the traversal could miss a violation of the height-balance invariant. To mitigate this risk, the script was cross-checked against a hand-crafted reference tree for small dimensions. **External validity** is limited to C implementations that follow the same AVL-based layout representation; alternative data structures (e.g., B-trees) may exhibit different rebalancing costs. the analysis assumes a deterministic compiler without aggressive inlining or vectorization that could alter the observable control-flow graph. Finally, the absence of runtime measurements means that cache-related latency effects on actual GPUs remain unquantified.

8 Conclusion

We presented a concise C library that formalizes layout manipulation for GPU kernels as a height-balanced search algebra. Formal invariants guarantee bijective offset translation and logarithmic update costs, while static AST analysis confirms the library's minimal code footprint. Future work will integrate the algebra with auto-tuning frameworks and extend the proof infrastructure to cover GPU-specific memory-coalescing properties.

References

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